INCH-POUND
MIL-DTL-24643/33G
27 November 2023
SUPERSEDING
MIL-DTL-24643/33F
1 October 2009

DETAIL SPECIFICATION SHEET

CABLE, ELECTRICAL, -4 DEGREES FAHRENHEIT (-20 DEGREES CELSIUS) TO +194 DEGREES FAHRENHEIT (+90 DEGREES CELSIUS), 600 VOLTS, TYPE LS2SWU

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-DTL-24643.

Construction (Watertight)

First	-	Copper conductor, tin-coated, 18 AWG, class B stranding (7 by 0.0152 inches) in accordance with ASTM B8.
Second	-	Separator may be used at manufacturer's option where required to provide free-stripping insulation.
Third	-	Thermoset insulation. Nominal wall thickness of 0.023 inch; wall thickness may vary from nominal as necessary, in order for the completed cable to meet the specified electrical requirements. Colored insulation, one white and one black conductor for each pair.
Fourth	-	Two conductors (one black and one white) cabled together with a nominal lay of 3 inches to form a pair.
Fifth	-	An optional binder.
Sixth	-	Braided shield of 34 or 36 AWG, tin-coated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.
Seventh	-	Shield insulation of two polyester tapes, outer tape sealed. An alternate shield insulation may consist of one polyester tape, plus an extruded jacket of a suitable transparent material; minimum average thickness of the transparent material 0.003 inch. Standard identification code shall be applied by method 2.
Eighth	-	The specified number of pairs (see table I) shall be cabled together in accordance with 3.4.4 of MIL-DTL-24643. Cabling sequence shall be consecutive, starting with pair no. 1, from the center outward. Fillers may be used as necessary to form a firm, well-rounded assembly.
Ninth	-	An optional binder.
Tenth	-	Cross-linked polyolefin jacket. See table I for thickness.

NOTE 1: Manufacturer's identification tape is required (see 3.4.9 of MIL-DTL-24643).

NOTE 2: For LS2SWU-1, omit seventh and eighth.

Distribution Statement A. Approved for public release: distribution is unlimited.

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TABLE I. Details.

Military part no. M24643/33	Type and size	Number of pairs	Cable jacket thickness min. avg. (inch)	Overall diameter (inches)		Hydrostatic leakage (cubic inches)
				min.	max.	
-01UN	LS2SWU-1	1	0.015	0.240	0.255	1.00
-02UN	LS2SWU-3	3	0.075	0.670	0.710	1.00
-03UN	LS2SWU-7	7	0.075	0.860	0.910	1.00
-04UN	LS2SWU-12	12	0.088	1.130	1.200	2.00
-05UN	LS2SWU-19	19	0.100	1.292	1.380	3.00
-06UN	LS2SWU-24	24	0.100	1.500	1.590	3.50
-07UN	LS2SWU-30	30	0.100	1.670	1.760	4.00
-08UN	LS2SWU-37	37	0.125	1.785	1.870	5.00
-09UN	LS2SWU-61	61	0.125	2.205	2.300	8.00

REQUIREMENTS:

Qualification required.

INSPECTION:

Basic electricals:

Conductor resistance (ohms/1,000 feet at 77 °F [25 °C], max.)	7.47
Voltage withstand (volts, root mean square, min.)	
Conductor to conductor	2,000
Conductor to shield	1,000
Component shield to component shield	500
Insulation resistance (megohms/1,000 feet, min.)	
Conductor to conductor	500
Conductor to shield	500
Component shield to component shield	100
Conductor continuity	No failure
Shield continuity	No failure
Jacket flaws (type LS2SWU-1 only)	No failure

Group A:

Visual and dimensional	No failure
Hydrostatic (open end) (leakage at 300 lb/in^2, 2 hours, in^3, max.)	See table I

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Capacitance	
Mutual capacitance (nominal at 1 kilohertz [kHz]) (picofarad [pF]/100 feet)	30
Capacitance unbalance (percent max.)	8
Characteristic impedance, at 1 megahertz (ohms)	75±5
Group B:	
Cold bending (cable)	No failure
Thermoset proof test (percent, max.)	
Insulation	50
Jacket (when tested at 392 °F [200 °C])	50
Drip (203±1.8 °F [95±1 °C])	No failure
Physicals (unaged)	
Insulation (extruded insulation only)	
Tensile strength (lb/in^2, min.)	700
Elongation (percent, min.)	150
Jacket (cable)	
Tensile strength (lb/in^2, min.)	1,300
Elongation (percent, min.)	160
Tear (lb/in thickness, min.)	35
Group C:	
Physicals (aged) (air oven)	
Insulation (extruded insulation only)	
Tensile strength (percent of unaged, min.)	75
Elongation (percent of unaged, min.)	75
Jacket (cable)	
Tensile strength (percent of unaged, min.)	60
Elongation (percent of unaged, min.)	60
Permanence of printing (jacket) (cycles, min.)	125
Heat distortion (percent of unaged, max.)	30
Cable fill (sealant) removability	No failure
Shrinkage	No failure
Shield (conformance to material, construction, and coverage)	No failure

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Surface transfer impedance (type LS2SWU-1 only)	
Milliohms per meter (max.)	Not applicable
Electromagnetic pulse (EMP) response (dB, min.)	Not applicable

Group D:

Flame propagation (cable)	No failure

Qualification inspection:

Qualification inspection shall include basic electricals; groups A, B, C, and D; plus the following:

Aging and compatibility (cable) (257±9 °F [125±5 °C])	No failure
Abrasion resistance (jacket) (scrapes, min.)	75
Acid gas equivalent (percent, max.)	
Jacket	2
Fillers	2
Insulation	18
Shield insulation	2
Halogen content (percent, max.)	
Jacket	0.2
Fillers	0.2
Insulation	0.2
Immersion (jacket)	
Tensile strength (percent of unaged, min.)	50
Elongation (percent of unaged, min.)	50
Smoke index (max.)	
Jacket	25
Fillers	45
Insulation	45
Toxicity index (max.)	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) – type A (hardness, min.)	80
Weathering (jacket)	No failure

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Electrical moisture absorption	No failure

UNIT ORDERING LENGTHS:

Type	Size	Feet (nominal)
LS2SWU	1 through 12	1,500
LS2SWU	19 and 24	1,000
LS2SWU	30 through 61	500

CHANGES FROM PREVIOUS ISSUE: Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodians:
Army – MI
Navy – SH

Preparing activity:
Navy – SH
(Project 6145-2013-049)

Review activities:
Army – AR, CR
Navy – CG, EC
DLA – CC

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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